CCP REPORT

(CT-175)

Mini Shopping Cart in C

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1.Project Title: Mini Shopping Cart

2.Objectives:

The objective of this project is to develop a console-based Mini Shopping Cart system in C language. It allows users to select categories, choose items, specify quantities, and calculate totals automatically with discounts applied based on the purchase amount.

3. Tools and Technologies:

- Programming Language: C
- Compiler: GCC (Dev-C++)
- Platform: Windows

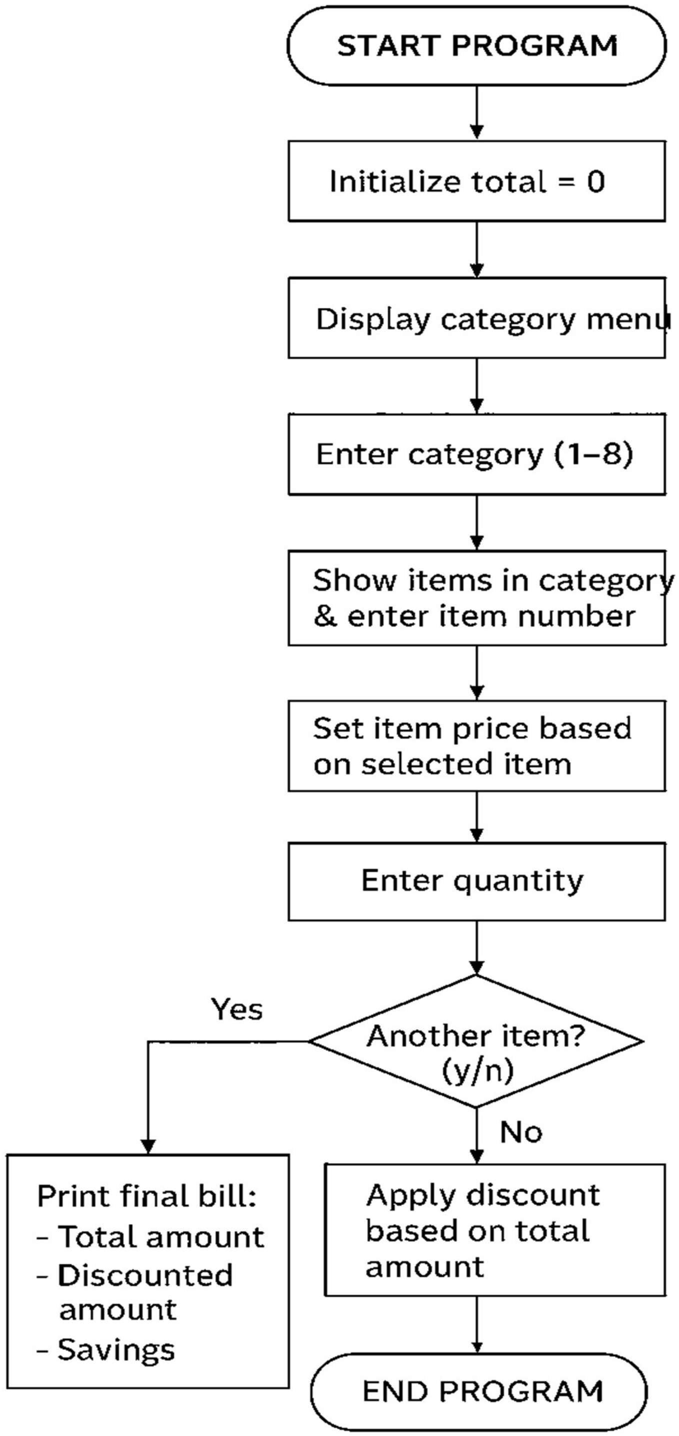
4. Project Description:

This Mini Shopping Cart program is a text-based system created in the C programming language. It mimics the behavior of a real shopping cart where users can select different product categories such as Dairy, Meat, and Fruits, choose items, and enter quantities. The program automatically calculates the subtotal, applies a discount according to the total, and displays the final bill. This project demonstrates practical usage of loops, conditions, switch cases, and user-defined functions in C.

5. Features:

- Category-wise item selection
- Dynamic total and subtotal calculation
- Discount system based on total purchase
- Final bill with total, discount, and savings
- Error handling for invalid selections

6.Flowchart:


START PROGRAM
Initialize total = 0
Display category menu
Enter category (1–8)
Show items in category & enter item number
Set item price based on selected item
Enter quantity
Another item? (y/n)
Yes
No
Print final bill:
- Total amount
- Discounted amount
- Savings
Apply discount based on total amount
END PROGRAM

7. Source Code:

Header files:

- <stdio.h>
- <string.h>

Functions:

- float discount_calc(float total);
- void bill(float total, float discountedTotal);
- void showCategories();

Variables:

- int category, item, quantity;
- float price = 0, itemTotal, total = 0, discountedTotal;
- char choice;

Conditional Statements:

- Switch statement
- If statement
- Nested if else statement

Iterative statement:

- Do while loop

Jump Statements:

- Return statement
- Continue statement
- Break statement

Function Body:

```
// Function to display the categories
void showCategories()
{
    printf("\n================= CATEGORIES =================");
    printf("\n1. Dairy");
    printf("\n2. Meat");
    printf("\n3. Cooking Stuff");
    printf("\n4. Ready Made");
    printf("\n5. Laundry");
    printf("\n6. Fruits & Vegetables");
    printf("\n7. Canned Stuff");
    printf("\n8. Bath");
    printf("\n=============================================\n");
}


// Function to apply discount
float discount_calc(float total)
{
    float discount = 0;
```

```
    if (total >= 10000)
        discount = 0.40 * total;
    else if (total >= 5000)
        discount = 0.25 * total;
    else if (total >= 3000)
        discount = 0.15 * total;

    return total - discount;
}


// Function to print the final bill
void bill(float total, float discountedTotal)
{
    printf("\n==========================================\n");
    printf("--------------- FINAL BILL ---------------\n");
    printf("==========================================\n");
    printf("Total Amount:        Rs. %.2f\n", total);
    printf("Discounted Amount:   Rs. %.2f\n", discountedTotal);
    printf("You Saved:           Rs. %.2f\n", total - discountedTotal);
    printf("------------------------------------------\n");
    printf("   Thank you for shopping with us!\n");
    printf("------------------------------------------\n");
}
```

Switch statement:

```
switch (category)
  {
      case 1: // Dairy section
          printf("\n1. Milk (Rs. 180 per litre)");
          printf("\n2. Cheese (Rs. 500 per pack)");
          printf("\n3. Butter (Rs. 350 per pack)");
          printf("\n4. Yoghurt (Rs. 250 per kg)");
          printf("\nChoose item: ");
          scanf("%d", &item);
          if (item == 1)
              price = 180;
          else if (item == 2)
              price = 500;
          else if (item == 3)
              price = 350;
          else if (item == 4)
              price = 250;
          else{
            printf("\nInvalid item selected! Please try again.\n");
            continue;}
          break;

      case 2: // Meat section
          printf("\n1. Chicken (Rs. 650 per kg)");
```

```
    printf("\n2. Beef (Rs. 900 per kg)");
    printf("\n3. Mutton (Rs. 1300 per kg)");
    printf("\n4. Fish (Rs. 2000 per kg)");
    printf("\nChoose item: ");
    scanf("%d", &item);
    if (item == 1)
        price = 650;
    else if (item == 2)
        price = 900;
    else if (item == 3)
        price = 1300;
    else if (item == 4)
        price = 2000;
    else{
        printf("\nInvalid item selected! Please try again.\n");
        continue;}
    break;

case 3: // Cooking stuff
    printf("\n1. Oil (Rs. 650 per litre)");
    printf("\n2. Flour (Rs. 180 per kg)");
    printf("\n3. Rice (Rs. 300 per kg)");
    printf("\n4. Salt (Rs.50 per pack)");
    printf("\n5. Sugar (Rs.100 per kg)");
    printf("\n6. Beans (Rs.200 per kg)");
    printf("\n7. Tea leaves (Rs.265 per pack)");
```

```
    printf("\nChoose item: ");
    scanf("%d", &item);
    if (item == 1)
        price = 650;
    else if (item == 2)
        price = 180;
    else if (item == 3)
        price = 300;
    else if (item == 4)
        price = 50;
    else if (item == 5)
        price = 100;
    else if (item == 6)
        price = 200;
    else if (item == 7)
        price = 265;
    else{
         printf("\nInvalid item selected! Please try again.\n");
          continue;}

    break;

case 4: // Ready made
    printf("\n1. Bread (Rs. 180 per pack)");
    printf("\n2. Biscuits (Rs. 120 per pack)");
    printf("\n3. Cake (Rs. 500 per cake)");
```

```
    printf("\nChoose item: ");
    scanf("%d", &item);
    if (item == 1)
        price = 180;
    else if (item == 2)
        price = 120;
    else if (item == 3)
        price = 500;
    else{
        printf("\nInvalid item selected! Please try again.\n");
        continue;}

    break;

case 5: // Laundry
    printf("\n1. Surf Excel (Rs. 550 per kg)");
    printf("\n2. Ariel (Rs. 600 per kg)");
    printf("\n3. Comfort (Rs. 250 per bottle)");
    printf("\nChoose item: ");
    scanf("%d", &item);
    if (item == 1)
        price = 550;
    else if (item == 2)
        price = 600;
    else if (item == 3)
        price = 250;
```

```
    else{
      printf("\nInvalid item selected! Please try again.\n");
      continue;}

    break;

case 6: // Fruits and Veggies
    printf("\n1. Apples (Rs. 300 per kg)");
    printf("\n2. Tomatoes (Rs. 150 per kg)");
    printf("\n3. Potatoes (Rs. 120 per kg)");
    printf("\nChoose item: ");
    scanf("%d", &item);
    if (item == 1)
        price = 300;
    else if (item == 2)
        price = 150;
    else if (item == 3)
        price = 120;
    else{
         printf("\nInvalid item selected! Please try again.\n");
      continue;}

    break;

case 7: // Canned stuff
    printf("\n1. Baked Beans (Rs. 250 per can)");
```

```
    printf("\n2. Tuna (Rs. 350 per can)");
    printf("\n3. Sweet Corn (Rs. 200 per can)");
    printf("\nChoose item: ");
    scanf("%d", &item);
    if (item == 1)
        price = 250;
    else if (item == 2)
        price = 350;
    else if (item == 3)
        price = 200;
    else{
        printf("\nInvalid item selected! Please try again.\n");
        continue;}

    break;

  case 8: // Bath
    printf("\n1. Soap (Rs. 180 per bar)");
    printf("\n2. Shampoo (Rs. 400 per bottle)");
    printf("\n3. Toothpaste (Rs. 250 per pack)");
    printf("\nChoose item: ");
    scanf("%d", &item);
    if (item == 1)
        price = 180;
    else if (item == 2)
        price = 400;
```

```
else if (item == 3)
    price = 250;
else{
    printf("\nInvalid item selected! Please try again.\n");
    continue;}
    break;
}
```

8. Testing and Results:

We tested the program under various input cases including:

- Normal purchase flow across multiple categories
- Invalid inputs and error handling
- Boundary cases for discount thresholds at Rs. 3000, 5000, and 10000

All tests passed successfully, and the output matched the expected results.

9. Future Enhancements:

- Add file handling to save and load shopping sessions
- Introduce arrays or structures to handle item data efficiently
- Add search, remove, or edit options in the cart
- Improve interface and implement advanced error handling

10. Conclusion:

Through this Mini Shopping Cart project, we learned how to apply fundamental C programming concepts to solve real-world problems. We gained hands-on experience in using loops, conditional statements, switch cases, and functions to design an interactive and user-friendly program. This project enhanced our problem-solving and logical thinking skills.